

PATTERN FORMATION PDES

So far, we have associated diffusion with a smoothing and homogenizing (eliminates chemical gradients and leads to uniform spatial distribution)

But it can have an opposite effect: fostering nonuniform chemical "patterns"
The following provides the mathematical basis for this claim:

Necessary conditions for pattern formation are:

- 1) Two or more chemical species
- 2) Different rates of diffusion
- 3) Chemical interactions

An appropriate combination of these factors can result in chemical patterns that arise as a destabilization of a uniform chemical distribution.

(A. Turing 1952 → Segel & Jackson 1972)

$C_1 = C_1(x, t)$ and $C_2 = C_2(x, t)$: conc. of chemicals 1 and 2

$$(*) \quad \begin{cases} \frac{\partial C_1}{\partial t} = R_1(C_1, C_2) + D_1 \frac{\partial^2 C_1}{\partial x^2} \\ \frac{\partial C_2}{\partial t} = R_2(C_1, C_2) + D_2 \frac{\partial^2 C_2}{\partial x^2} \end{cases}$$

- ① To underscore the role of diffusion in pattern formation, we assume that in its absence solutions are well mixed → chemical system arrives at a spatially uniform steady state: $\bar{C}_1, \bar{C}_2$.

This means:

$$\frac{\partial \bar{C}_1}{\partial t} = 0 \quad \& \quad \frac{\partial^2 \bar{C}_1}{\partial x^2} = 0 \quad (\text{and similarly for } C_2)$$

We therefore conclude:

$$R_1(\bar{C}_1, \bar{C}_2) = 0$$

$$R_2(\bar{C}_1, \bar{C}_2) = 0$$

- ② Are these solutions stable?

Consider small inhomogeneous perturbations:

$$C_1'(x, t) = C_1(x, t) - \bar{C}_1$$

$$C_2'(x, t) = C_2(x, t) - \bar{C}_2$$

Linearize (*) about the homogeneous steady state, to obtain:

$$\begin{cases} \frac{\partial C_1'}{\partial t} = a_{11} C_1' + a_{12} C_2' + D_1 \frac{\partial^2 C_1'}{\partial x^2} \\ \frac{\partial C_2'}{\partial t} = a_{21} C_1' + a_{22} C_2' + D_2 \frac{\partial^2 C_2'}{\partial x^2} \end{cases}$$

where $a_{11} = \partial R_1 / \partial C_1 |_{\bar{C}_1, \bar{C}_2}$ $a_{12} = \partial R_1 / \partial C_2 |_{\bar{C}_1, \bar{C}_2}$

$a_{21} = \partial R_2 / \partial C_1 |_{\bar{C}_1, \bar{C}_2}$ $a_{22} = \partial R_2 / \partial C_2 |_{\bar{C}_1, \bar{C}_2}$

(**) Alternatively: $\vec{C}'_t = M \vec{C}' + D \vec{C}'_{xx}$

$$\begin{bmatrix} C_1' \\ C_2' \end{bmatrix}_t = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} C_1' \\ C_2' \end{bmatrix} + \begin{bmatrix} D_1 & 0 \\ 0 & D_2 \end{bmatrix} \begin{bmatrix} C_1' \\ C_2' \end{bmatrix}_{xx}$$

Recall, we have solved such equations using separation of variables:

(v) $\vec{C}' = \begin{bmatrix} C_1' \\ C_2' \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} \cos(qx) e^{\sigma t}$

$\downarrow$ amplitude at time $t=0$ $\downarrow$ growth rate $\downarrow$ wave number

Plug into (**) to determine stability of these solutions:

$$\alpha_1 \sigma = a_{11} \alpha_1 + a_{12} \alpha_2 - D_1 q^2 \alpha_1$$

$$\alpha_2 \sigma = a_{21} \alpha_1 + a_{22} \alpha_2 - D_2 q^2 \alpha_2$$

(v) is a solution to (**) provided these equations hold!

$$\alpha_1 (\sigma - a_{11} + D_1 q^2) + \alpha_2 (-a_{12}) = 0$$

$$\alpha_1 (-a_{21}) + (\sigma - a_{22} + D_2 q^2) \alpha_2 = 0$$

Trivial solution: $\alpha_1 = 0, \alpha_2 = 0$. (since RHS = 0)

(Actually unique soln. unless the eqns. are redundant!)

But we don't want this solution (perturbations are absent for all time)

Thus: $\det = 0$

③ Can we find growing perturbations due to diffusion?

"characteristic eqn"

$$\det \begin{pmatrix} \sigma - a_{11} + D_1 q^2 & -a_{12} \\ -a_{21} & \sigma - a_{22} + D_2 q^2 \end{pmatrix} = 0 \Rightarrow \sigma^2 + \sigma[-a_{22} + D_2 q^2 - a_{11} + D_1 q^2] + [(a_{11} - D_1 q^2)(a_{22} - D_2 q^2) - a_{12}a_{21}] = 0.$$

can we find a wave number q that is unstable (growing waviness), σ .

A. Conditions for stability of chemicals (in the absence of diffusion)

$$D_1 = D_2 = 0$$

$$\Rightarrow \sigma^2 - \sigma(a_{11} + a_{22}) + (a_{11}a_{22} - a_{12}a_{21}) = 0.$$

$\text{Re}(\sigma) < 0$ (negative $\rightarrow$ decaying amplitude)

$$\sigma = \frac{(a_{11} + a_{22}) \pm \sqrt{(a_{11} + a_{22})^2 - 4(a_{11}a_{22} - a_{12}a_{21})}}{2}$$

$$(1) a_{11}a_{22} - a_{12}a_{21} > 0$$

$$(2) a_{11} + a_{22} < 0$$

B. Conditions for stability of chemicals (with diffusion)

$$(1) (a_{11} - D_1 q^2)(a_{22} - D_2 q^2) - a_{12}a_{21} > 0$$

$$(2) (a_{11} + a_{22} - D_2 q^2 - D_1 q^2) < 0$$

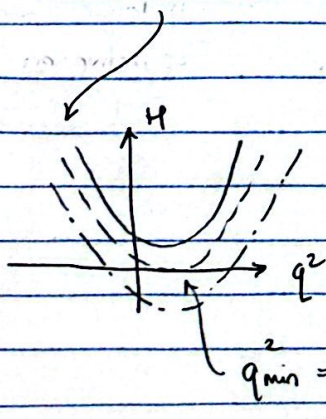
Notice that (2) is automatically satisfied if $a_{11} + a_{22} < 0$!

(since $D_1, D_2 > 0$)

Therefore, for diffusive instability to arise, (1) must be violated.

$$H = D_1 D_2 (q^2)^2 - (D_1 a_{22} + D_2 a_{11}) (q^2) + (a_{11}a_{22} - a_{12}a_{21}) < 0$$

for diffusive instability



Alternatively :

$$(3) a_{11}D_2 + a_{22}D_1 > 2\sqrt{D_1 D_2} \sqrt{a_{11}a_{22} - a_{12}a_{21}}$$

$$q_{\min}^2 = \frac{1}{2} \left(\frac{a_{22}}{D_2} + \frac{a_{11}}{D_1} \right) \rightarrow$$

Physical Explanation: (Thanks to Segel & Jackson 1972)

CONDITIONS FOR DIFFUSIVE INSTABILITY:

(1) $a_{11} a_{22} - a_{12} a_{21} > 0$

(2) $a_{11} + a_{22} < 0$

(3) $a_{11} D_2 + a_{22} D_1 > 2\sqrt{D_1 D_2} \sqrt{a_{11} a_{22} - a_{12} a_{21}} > 0$

(2) says at least one, a_{11} or a_{22} , is negative. WLOG a_{22} is negative

$$a_{22} = \frac{\partial R_2}{\partial C_2} < 0$$

rate of production
of chem. 2 is inhibited by presence of chem 2

"INHIBITOR"

(3) says $a_{11} D_2 + a_{22} D_1 > 0 \Rightarrow a_{11}$ is positive

$$a_{11} = \frac{\partial R_1}{\partial C_1} > 0$$

chem. 1 promotes or activates
its own production

"ACTIVATOR"

Above says $a_{11} a_{22} < 0$. Then if $a_{11} a_{22} - a_{12} a_{21} > 0 \Rightarrow a_{12} a_{21} < 0$.

$$M = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

(A) $a_{12} < 0, a_{21} > 0$

$$M = \begin{pmatrix} + & - \\ + & - \end{pmatrix}$$

"activator - inhibitor"

(B) $a_{12} > 0, a_{21} < 0$

$$M = \begin{pmatrix} + & + \\ - & - \end{pmatrix}$$

"positive feedback"
(substrate depletion)

(3) also says $a_{11} + a_{22} \frac{D_1}{D_2} > 0 \Rightarrow D_1 \neq D_2$ (otherwise contradicts (2))

lastly, define $\tau_1 = 1/|a_{11}|$ and $\tau_2 = 1/|a_{22}|$

(3) yields: $\sqrt{D_1 \tau_1} < \sqrt{D_2 \tau_2}$

range
of activation range
of inhibition

Range of inhibition $\sqrt{D_2 \tau_2}$
is larger than range
of activation $\sqrt{D_1 \tau_1}$